REVISION NOTES

Plus Two Higher Secondary (Class 12) Chemistry - Quick Revision Formula Book & Reactions

Solutions, Electrochemistry, Chemical Kinetics, Coordination Compounds & Organic Named Reactions

Prescribed Institution / Board:	Directorate of Higher Secondary Education & CBSE Academic Wing
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EXECUTIVE OVERVIEW & PEDAGOGIC PURPOSE

This academic resource has been officially compiled and vetted for students enrolled under the approved curriculum. It contains exhaustive conceptual formulations, worked-out illustrations, previous examination evaluation schemes, and standard pedagogical solutions aligned with the 2026-2027 state academic syllabus.

Physical Chemistry: Solutions & Electrochemistry

1.1 Colligative Properties & Raoults Law

Relative Lowering of Vapour Pressure: (P0 - P) / P0 = x_solute = (w2 * M1) / (M2 * w1)

Elevation in Boiling Point: Î”Tb = Kb * m, where m is molality.

Depression in Freezing Point: Î”Tf = Kf * m

Osmotic Pressure: Î = C * R * T = (n / V) * R * T

Van t Hoff Factor (i): i = Normal Molar Mass / Abnormal Molar Mass = (1 - Î± + nÎ±) for dissociation.

1.2 Electrochemistry & Nernst Equation

Nernst Equation: E_cell = E0_cell - (0.0591 / n) * log10([Products] / [Reactants]) at 298 Kelvin.

Kohlrauschs Law: Limiting molar conductivity of an electrolyte is represented as the sum of individual contributions of the anion and cation: Î›0_m = Î½+ * Î»0+ + Î½- * Î»0-.

Faradays Laws of Electrolysis: m = Z * I * t, where electrochemical equivalent Z = Equivalent Weight / 96500.

Inorganic & Organic Chemistry: Reactions & Formulations

2.1 Coordination Compounds & Crystal Field Theory (CFT)

IUPAC Nomenclature rules: Cation named first, ligands listed alphabetically before central metal ion.

Crystal Field Splitting in Octahedral complexes: d-orbitals split into lower t2g (triply degenerate) and upper eg (doubly degenerate) sets. If Î”o > P (pairing energy), low spin complex forms.

2.2 Crucial Organic Named Reactions

Aldol Condensation: Aldehydes having Î±-hydrogen react with dilute NaOH to yield Î²-hydroxy aldehydes.

Cannizzaro Reaction: Aldehydes lacking Î±-hydrogen (Formaldehyde, Benzaldehyde) undergo self oxidation-reduction with concentrated alkali to yield alcohol and salt of carboxylic acid.

Sandmeyers Reaction: Benzene diazonium chloride reacts with CuCl/HCl or CuBr/HBr to yield Chlorobenzene or Bromobenzene.